Introduction to Analysis II

Lecture notes, Chapters 6–10

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Chapter 6. Sequences and Series of Functions

When can limits of functions be exchanged with continuity, derivatives, integrals and other limits? The answer, in most cases, is uniform convergence.

6.1 Uniform convergence

Let $f_n, f : X \to \mathbb{R}^m$ ($X \subset \mathbb{R}^k$). $f_n \to f$ **pointwise** if $f_n(x) \to f(x)$ for each x, and **uniformly** if $\|f_n - f\|_\infty = \sup_{x\in X}\|f_n(x) - f(x)\| \to 0$, i.e. N can be chosen independently of x.

Example 6.1

(a) x^n on $[0,1]$ converges pointwise to the function equal to 0 on $[0,1)$ and 1 at 1: continuous functions with a discontinuous limit, so the convergence is not uniform. On $[0,r]$ with $r<1$ it is uniform.
(b) $f_n(x) = nx(1-x^2)^n$ on $[0,1]$ tends to 0 pointwise, but $\int_0^1 f_n = \frac{n}{2n+2} \to \frac{1}{2}$.

Theorem 6.1

A uniform limit of continuous functions is continuous.

Proof. $\|f(x) - f(x_0)\| \le \|f(x) - f_n(x)\| + \|f_n(x) - f_n(x_0)\| + \|f_n(x_0) - f(x_0)\|$: choose n with $\|f_n - f\|_\infty < \epsilon$, then δ from the continuity of f_n at x_0 (the "$\epsilon/3$ argument"). ■

Theorem 6.2 (Uniform Cauchy criterion)

(f_n) converges uniformly on X iff for every $\epsilon > 0$ there is N with $\|f_n - f_m\|_\infty < \epsilon$ for $m, n \ge N$.

Proof. Sufficiency: for each x the sequence is Cauchy, so it converges to some $f(x)$; letting $m \to \infty$ in $\|f_n(x) - f_m(x)\| < \epsilon$ gives $\|f_n - f\|_\infty \le \epsilon$ for $n \ge N$. ■

Corollary 6.3 (Weierstrass M-test)

If $|f_n(x)| \le M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly on X.

Example 6.2

(a) $\sum \frac{\sin nx}{n^2}$ converges uniformly on $\mathbb{R}$ to a continuous function.
(b) (A continuous nowhere differentiable function.) Let $\phi(x)$ be the distance from x to the nearest integer and $f(x) = \sum_{k\ge 0}(3/4)^k \phi(4^k x)$. The M-test gives continuity. At any x, choosing $h_m = \pm\frac{1}{2}4^{-m}$ so that no integer lies between $4^m x$ and $4^m(x + h_m)$ makes $|f(x + h_m) - f(x)|/|h_m| \ge 3^m - \sum_{k<m} 3^k = \frac{3^m+1}{2} \to \infty$, so f is nowhere differentiable.

6.2 Differentiation and integration of limits

Theorem 6.4

Let f_n be differentiable on $[a,b]$, $(f_n(x_0))$ converge for some x_0, and (f_n') converge uniformly. Then (f_n) converges uniformly to a differentiable f with $f' = \lim f_n'$.

Proof. The mean value theorem applied to $f_n - f_m$ gives $|(f_n - f_m)(x) - (f_n - f_m)(t)| \le \|f_n' - f_m'\|_\infty |x - t|$. With $t = x_0$ this makes (f_n) uniformly Cauchy. Fix x and put $\phi_n(t) = \frac{f_n(t)-f_n(x)}{t-x}$, $\phi(t) = \frac{f(t)-f(x)}{t-x}$; the same estimate shows $\phi_n \to \phi$ uniformly on $t \neq x$, and $\lim_{t\to x} \phi_n(t) = f_n'(x)$. Exchanging the two limits (allowed by uniformity, as in Theorem 6.1) gives $f'(x) = \lim f_n'(x)$. ■

Uniform convergence of f_n alone is not enough: $\frac{\sin nx}{\sqrt{n}} \to 0$ uniformly but $\sqrt{n}\cos nx$ diverges.

Proposition 6.5

If $f_n \in \mathcal{R}[a,b]$ and $f_n \to f$ uniformly, then $f \in \mathcal{R}[a,b]$ and $\int_a^b f_n \to \int_a^b f$.

Proof. With $\epsilon_n = \|f_n - f\|_\infty$, $f_n - \epsilon_n \le f \le f_n + \epsilon_n$, so $\overline{\int} f - \underline{\int} f \le 2\epsilon_n(b-a) \to 0$ and $|\int f - \int f_n| \le \epsilon_n(b-a)$. ■

Hence $\sum f_n$ may be integrated term by term when it converges uniformly. Example 6.1(b) shows what goes wrong without uniformity; Chapter 10 gives far more flexible conditions (dominated and monotone convergence).

6.3 Double sequences and double series

Proposition 6.6 (Exchanging limits)

Let (a_{mn}) be a double sequence such that $a_{mn} \to b_m$ as $n \to \infty$ uniformly in m, and $a_{mn} \to c_n$ as $m \to \infty$ for each n. Then $\lim_m b_m$ and $\lim_n c_n$ exist and are equal.

Proof. Uniformity in m makes (c_n) Cauchy: $|a_{mn} - a_{mn'}| < \epsilon$ for $n, n' \ge N$ and all m; let $m \to \infty$. With $c = \lim c_n$, $|b_m - c| \le |b_m - a_{mn}| + |a_{mn} - c_n| + |c_n - c|$; fix n large to control the first and third terms, then let $m \to \infty$. ■

This is the abstract form of Theorem 6.1 (take $a_{mn} = f_n(x_m)$). For double series the cleanest result concerns nonnegative terms.

Proposition 6.7 (Double series)

If $a_{mn} \ge 0$, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$ (possibly $+\infty$), and both equal the sum over $\mathbb{N}^2$ in any order. If $\sum_m \sum_n |a_{mn}| < \infty$, the same holds for real or complex a_{mn}.

Proof. For finite M, N, $\sum_{m\le M} \sum_{n\le N} a_{mn} \le \sum_n \sum_m a_{mn}$; let $N \to \infty$, then $M \to \infty$; and symmetrically. Every finite partial sum over $\mathbb{N}^2$ is bounded by either iterated sum, and conversely. For absolutely summable families, split into positive and negative (real and imaginary) parts. ■

Without absolute convergence the iterated sums can differ: with $a_{mm} = 1$, $a_{m,m+1} = -1$ and 0 otherwise, $\sum_m \sum_n = 0$ but $\sum_n \sum_m = 1$.

6.4 Power series

Theorem 6.8 (Cauchy–Hadamard)

Let $A = \limsup|a_n|^{1/n}$ and $R = 1/A$ ($R = \infty$ if $A = 0$, $R = 0$ if $A = \infty$). Then $\sum a_n x^n$ converges absolutely for $|x| < R$, diverges for $|x| > R$, and converges uniformly on $[-r, r]$ for every $r < R$.

Proof. Apply the root test: $\limsup|a_n x^n|^{1/n} = A|x|$. For $|x| \le r < R$, the M-test with $M_n = |a_n| r_1^n$, $r < r_1 < R$, gives uniformity. ∎

So $f(x) = \sum a_n x^n$ is continuous on $(-R, R)$. Since $\limsup(n|a_n|)^{1/n} = A$, the differentiated series has the same radius, and Theorem 6.4 gives:

Theorem 6.9

$f(x) = \sum a_n x^n$ is infinitely differentiable on $(-R, R)$ with $f'(x) = \sum n a_n x^{n-1}$; in particular $a_n = f^{(n)}(0)/n!$. Moreover, for $|c| < R$, $f(x) = \sum_k \frac{f^{(k)}(c)}{k!}(x-c)^k$ for $|x - c| < R - |c|$.

Proof. The re-expansion: $\sum_n a_n(c + (x-c))^n = \sum_n \sum_k \binom{n}{k} a_n c^{n-k}(x-c)^k$, and the double series converges absolutely when $|c| + |x - c| < R$, so Proposition 6.7 allows the order to be exchanged. ∎

Proposition 6.10 (Cauchy product)

If $\sum a_n = A$ converges absolutely and $\sum b_n = B$ converges, then $\sum c_n$, $c_n = \sum_{k=0}^{n} a_k b_{n-k}$, converges to AB (Mertens).

Proof. With $B_n = \sum_{k \le n} b_k$ and $\beta_n = B_n - B \to 0$, $\sum_{k \le n} c_k = \sum_{k \le n} a_k B_{n-k} = A_n B + \sum_{k \le n} a_k \beta_{n-k}$, and the last sum tends to 0 by splitting at a large index and using $\sum|a_k| < \infty$. ∎

Theorem 6.11 (Abel)

If $\sum a_n$ converges, then $\sum a_n x^n$ converges uniformly on $[0, 1]$; hence $\lim_{x \to 1-} \sum a_n x^n = \sum a_n$.

Proof. Let $r_n = \sum_{k > n} a_k \to 0$. Summation by parts gives $\sum_{k=m}^{n} a_k x^k = r_{m-1} x^m - r_n x^n - \sum_{k=m}^{n-1} r_k (x^k - x^{k+1})$, which is at most $3 \sup_{k \ge m-1}|r_k|$ for $0 \le x \le 1$, because $\sum(x^k - x^{k+1}) \le 1$. ∎

Example 6.3

(a) $\log(1 + x) = \sum_{n \ge 1} (-1)^{n+1} x^n / n$ on $(-1, 1)$ (integrate $1/(1+x)$ term by term); Abel's theorem gives $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \cdots$. Similarly $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ from arctan.
(b) As a consequence of Abel's theorem, if $\sum a_n$, $\sum b_n$ and their Cauchy product $\sum c_n$ all converge, then $\sum c_n = (\sum a_n)(\sum b_n)$: apply Abel's theorem to $\sum c_n x^n = (\sum a_n x^n)(\sum b_n x^n)$, valid for $|x| < 1$.
(c) The converse of Abel's theorem fails: $\sum(-1)^n x^n \to \frac{1}{2}$ as $x \to 1-$ but $\sum(-1)^n$ diverges.

6.5 Trigonometric series

For complex vectors we use $\langle z, w \rangle = \sum z_i \overline{w_i}$; the Cauchy–Schwarz inequality $|\langle z, w \rangle| \le \|z\|\|w\|$ holds in any complex inner product space, with the same proof (minimize $\|z - tw\|^2$ over complex t).

Theorem 6.12 (Dirichlet's test)

Let f_n be complex functions on X whose partial sums $F_n = \sum_{k\le n} f_k$ are uniformly bounded, and let g_n be real with $g_1 \ge g_2 \ge \cdots \ge 0$ and $g_n \to 0$ uniformly. Then $\sum f_n g_n$ converges uniformly on X.

Proof. Summation by parts: $\sum_{k=m}^{n} f_k g_k = F_n g_n - F_{m-1} g_m + \sum_{k=m}^{n-1} F_k (g_k - g_{k+1})$. If $|F_k| \le M$, the absolute value is at most $M\{g_n + g_m + (g_m - g_n)\} = 2M g_m \to 0$ uniformly; apply the uniform Cauchy criterion. ∎

In particular, if $\sum a_n$ has bounded partial sums and $b_n \downarrow 0$, then $\sum a_n b_n$ converges (this contains the alternating series test).

Theorem 6.13

If $b_n \downarrow 0$, then $\sum b_n e^{inx}$, $\sum b_n \cos nx$ and $\sum b_n \sin nx$ converge uniformly on $\{\delta \le |x| \le \pi\}$ for every $\delta \in (0, \pi)$. In particular $\sum b_n \sin nx$ converges for every x.

Proof. $\left|\sum_{k\le n} e^{ikx}\right| = \left|\frac{e^{i(n+1)x} - e^{ix}}{e^{ix} - 1}\right| \le \frac{1}{|\sin(x/2)|} \le \frac{1}{\sin(\delta/2)}$; apply Dirichlet's test. At $x \in \pi\mathbb{Z}$ the sine series is 0. ∎

More generally the conclusion holds if $b_n \to 0$ and $\sum |b_n - b_{n-1}| < \infty$ (the same proof). For instance $\sum \frac{\sin nx}{n}$ converges everywhere (to $\frac{\pi - x}{2}$ on $(0, 2\pi)$, Chapter 9), but not uniformly near 0.

Theorem 6.14 (Abel's test)

If $\sum f_n$ converges uniformly on X and (g_n) are real functions, uniformly bounded and monotone in n for each x, then $\sum f_n g_n$ converges uniformly.

Proof. Apply summation by parts with the tails $r_n = \sum_{k>n} f_k$, which are uniformly small, as in the proof of Theorem 6.11. ∎

Chapter 7. Function Spaces

It is fruitful to regard functions as points of a vector space with a norm. Convergence of functions then becomes convergence of points, and questions such as "does every Cauchy sequence converge?" and "which sets are compact?" make sense.

7.1 The space of continuous functions

A **norm** on a real or complex vector space V is a function $\|\cdot\|$ with $\|v\| \geq 0$ (equality iff $v = 0$), $\|cv\| = |c|\|v\|$ and $\|v + w\| \leq \|v\| + \|w\|$. A normed space is **complete** (a **Banach space**) if every Cauchy sequence converges. For compact $X \subset \mathbb{R}^n$, $C(X)$ denotes the continuous functions $X \to \mathbb{R}$ (or $\mathbb{C}$) with $\|f\|_\infty = \max_X |f|$; convergence in this norm is uniform convergence.

Theorem 7.1

$C(X)$ is complete.

Proof. A Cauchy sequence converges uniformly by Theorem 6.2, and its limit is continuous by Theorem 6.1. ∎

Closed bounded subsets of $C(X)$ need not be compact: $f_n(x) = x^n$ in $C[0,1]$ is bounded but has no uniformly convergent subsequence. The missing ingredient is equicontinuity.

Definition (Equicontinuity)

A family $\mathcal{F} \subset C(X)$ is **equicontinuous** if for every $\epsilon > 0$ there is $\delta > 0$ such that $|f(x) - f(y)| < \epsilon$ whenever $\|x - y\| < \delta$ and $f \in \mathcal{F}$.

A uniformly convergent sequence in $C(X)$ is equicontinuous (X compact): use uniform continuity of finitely many f_n and the uniform closeness of the rest to the limit. A family with uniformly bounded derivatives (or a common Lipschitz constant) is equicontinuous.

Theorem 7.2 (Arzelà–Ascoli)

If X is compact and $\mathcal{F} \subset C(X)$ is bounded and equicontinuous, then every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.

Proof. Let $E = \{e_1, e_2, \dots\}$ be a countable dense subset of X (points with rational coordinates near X, suitably chosen). Since $(f_n(e_1))$ is bounded, a subsequence converges at e_1; from it extract a subsequence converging at e_2; and so on. The diagonal sequence g_k (the k-th function of the k-th subsequence) converges at every e_j. Given ϵ, take δ from equicontinuity; by compactness finitely many $e_{j_1}, \dots, e_{j_p}$ have every $x \in X$ within δ of one of them. Then $|g_k(x) - g_l(x)| \leq |g_k(x) - g_k(e_j)| + |g_k(e_j) - g_l(e_j)| + |g_l(e_j) - g_l(x)| < 3\epsilon$ for k, l large, uniformly in x. ∎

Hence a closed, bounded, equicontinuous subset of $C(X)$ is compact. A typical application: solutions of $y' = F(t, y)$ with continuous F exist (Peano's theorem), because approximate solutions form an equicontinuous family.

Theorem 7.3 (Weierstrass approximation)

Every continuous $f : [a, b] \to \mathbb{R}$ is a uniform limit of polynomials.

Proof. Rescale to $[0,1]$. The Bernstein polynomials $B_n f(x) = \sum_{k=0}^{n} f(\tfrac{k}{n}) \binom{n}{k} x^k (1-x)^{n-k}$ work: with $p_k(x) = \binom{n}{k} x^k (1-x)^{n-k}$, $\sum p_k = 1$, $\sum \tfrac{k}{n} p_k = x$, $\sum (\tfrac{k}{n} - x)^2 p_k = \tfrac{x(1-x)}{n} \leq \tfrac{1}{4n}$ (binomial mean

and variance). Given ϵ, take δ from uniform continuity and $M = \|f\|_\infty$; then

$$|B_n f(x) - f(x)| \le \sum_{|k/n - x| < \delta} \epsilon p_k + \sum_{|k/n-x|\ge\delta} 2M p_k \le \epsilon + \frac{2M}{4n\delta^2} < 2\epsilon$$

for large n, by Chebyshev's inequality. ∎

So polynomials are dense in $C[a,b]$; since polynomials with rational coefficients are dense too, $C[a,b]$ has a countable dense set (it is *separable*).

7.2 Sequence spaces

For sequences $a = (a(n))$ define

$$\|a\|_1 = \sum |a(n)|, \qquad \|a\|_2 = \Big(\sum |a(n)|^2\Big)^{1/2}, \qquad \|a\|_\infty = \sup |a(n)|,$$

and $\ell^1, \ell^2, \ell^\infty$ as the spaces where these are finite; c_0 is the space of sequences tending to 0 (with $\|\cdot\|_\infty$). Then $\ell^1 \subsetneq \ell^2 \subsetneq c_0 \subsetneq \ell^\infty$. The triangle inequality in ℓ^2 comes from the Cauchy–Schwarz inequality $\sum |a(n)b(n)| \le \|a\|_2 \|b\|_2$, and $\langle a, b\rangle = \sum a(n)\overline{b(n)}$ is an inner product.

Theorem 7.4

ℓ^1, ℓ^2, ℓ^∞ and c_0 are complete.

Proof. For ℓ^2: if (a_m) is Cauchy, each coordinate sequence $(a_m(n))_m$ is Cauchy, with limit $a(n)$. For any N, $\sum_{n\le N} |a_m(n) - a_k(n)|^2 < \epsilon^2$ for $m, k \ge M$; let $k \to \infty$ and then $N \to \infty$ to get $\|a_m - a\|_2 \le \epsilon$, which also shows $a \in \ell^2$. The other cases are identical. ∎

7.3 Improper integrals

If $f \in \mathcal{R}[a,c]$ for every $c < b$ (where $b \le \infty$ or f is unbounded near b), the **improper integral** is $\int_a^b f = \lim_{c\to b-} \int_a^c f$ when the limit exists; similarly at the left endpoint or at interior singular points (each separately).

Proposition 7.5 (Comparison)

If $0 \le f \le g$ and $\int_I g$ converges, so does $\int_I f$. More generally, if $|f| \le g$ and $\int_I g < \infty$, then $\int_I f$ converges (absolute convergence implies convergence).

Proof. For $f \ge 0$, $c \mapsto \int_a^c f$ is increasing and bounded. In general use the Cauchy criterion: $|\int_c^{c'} f| \le \int_c^{c'} g$. ∎

Proposition 7.6 (Integral test)

If $f : [1,\infty) \to (0,\infty)$ is decreasing, then $\sum f(n)$ and $\int_1^\infty f$ converge or diverge together; indeed $\sum_{n=1}^N f(n) - \int_1^N f$ decreases to a finite limit.

Proof. $f(n+1) \le \int_n^{n+1} f \le f(n)$. ∎

This gives $\sum n^{-p} < \infty$ iff $p > 1$, and Euler's constant $\gamma = \lim(\sum_{n\le N} \frac{1}{n} - \log N)$.

Example 7.1

(a) $\int_0^\infty \frac{\sin x}{x}dx$ converges (integrate by parts on $[1,c]$: $\int_1^c \frac{\sin x}{x} = \cos 1 - \frac{\cos c}{c} - \int_1^c \frac{\cos x}{x^2}$), but not absolutely: $\int_{k\pi}^{(k+1)\pi} |\sin x|/x \geq \frac{2}{(k+1)\pi}$.
(b) $\int_0^1 x^{-p}dx < \infty$ iff $p < 1$; $\int_1^\infty x^{-p}dx < \infty$ iff $p > 1$.

Theorem 7.7

If $\int_I |f| < \infty$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $\int_J |f| < \epsilon$ for every subinterval J of length less than δ (absolute continuity of the integral), and there is a compact interval $K \subset I$ with $\int_{I\setminus K} |f| < \epsilon$.

Proof. The second claim is the definition of convergence at the endpoints. For the first, choose K as in the second claim with bound $\epsilon/2$ and let $B = \sup_K |f|$ (f is bounded on compact subintervals of its domain of integrability); then $\int_J |f| \leq B|J| + \epsilon/2$. ■

7.4 Spaces of integrable functions

On an interval I let $\mathcal{R}^1(I)$ be the functions that are (properly or improperly) Riemann integrable together with $|f|$, and $\mathcal{R}^2(I)$ those with $\int_I |f|^2 < \infty$; set

$$\|f\|_1 = \int_I |f|, \qquad \|f\|_2 = \Big(\int_I |f|^2\Big)^{1/2}.$$

These are seminorms: $\|f\| = 0$ does not force $f = 0$ (change f at one point). Identifying functions with $\|f - g\| = 0$ gives normed spaces, and $\langle f, g\rangle = \int_I f\overline{g}$ is an inner product on $\mathcal{R}^2(I)$.

Proposition 7.8

A normed space is complete iff every absolutely convergent series $\sum v_k$ ($\sum \|v_k\| < \infty$) converges.

Proof. If complete, the partial sums are Cauchy. Conversely, given a Cauchy sequence choose $n_1 < n_2 < \dots$ with $\|v_{n_{k+1}} - v_{n_k}\| < 2^{-k}$; the telescoping series converges, so a subsequence converges, and then the whole Cauchy sequence converges. ■

Example 7.2

$\mathcal{R}^1[0,1]$ is not complete. Let $g(x) = |x|^{-1/2}$ for $0 < |x| \leq 1$ (and $g(0) = 0$), so $\int_{-1}^1 g = 4$, let $r_1, r_2, \dots$ enumerate the rationals in $[0,1]$, and $f_n(x) = \sum_{k\leq n} 2^{-k} g(x - r_k)$. Each $f_n \in \mathcal{R}^1[0,1]$ (finitely many integrable singularities) and $\|f_n - f_m\|_1 \leq 4\sum_{k>m} 2^{-k} \to 0$. If some $f \in \mathcal{R}^1[0,1]$ were the limit, f would have to be unbounded near every rational, i.e. on every subinterval, so f could not be Riemann integrable, properly or improperly, on any subinterval. This defect is what the Lebesgue integral repairs (Chapter 10).

Nevertheless nice functions are dense:

Theorem 7.9

For $f \in \mathcal{R}^2(I)$ (resp. $\mathcal{R}^1(I)$) and $\epsilon > 0$ there is a continuous g vanishing outside a compact subinterval with $\|f - g\|_2 < \epsilon$ (resp. $\|f - g\|_1 < \epsilon$). Step functions may be used instead of continuous ones.

Proof. First truncate to a compact subinterval K (Theorem 7.7) and to a bounded function, losing at most ϵ. For bounded $f \in \mathcal{R}(K)$ choose a partition with $U(f,P) - L(f,P) < \epsilon$; the step function s equal to m_i on each piece satisfies $0 \le f - s$ and $\int(f-s) < \epsilon$, hence $\|f-s\|_2^2 \le 2\sup|f| \cdot \epsilon$. Finally replace each jump of s by a steep linear piece, changing the integral by an arbitrarily small amount. ■

Corollary 7.10 (Continuity of translation)

For $f \in \mathcal{R}^p[a,b]$ $(p = 1, 2)$, extended by 0, $\lim_{h\to 0}\|f(\cdot + h) - f\|_p = 0$.

Proof. True for continuous g with compact support by uniform continuity; approximate f by such g and use $\|f(\cdot+h) - f\| \le 2\|f - g\| + \|g(\cdot + h) - g\|$. ■

Chapter 8. Functions Defined by Integrals

Many important functions are defined as integrals depending on a parameter, $\phi(x) = \int f(x,t)\,dt$. We study when such ϕ is continuous or differentiable, and then meet the gamma function and integral transforms.

8.1 Continuity and differentiation under the integral sign

Proposition 8.1

If f is continuous on $[a,b]\times[c,d]$, then $\phi(x)=\int_c^d f(x,t)dt$ is continuous on $[a,b]$. If moreover $\partial_x f$ exists and is continuous on $[a,b]\times[c,d]$, then ϕ is C^1 and $\phi'(x)=\int_c^d \partial_x f(x,t)dt$.

Proof. f is uniformly continuous on the compact rectangle, so $|f(x,t)-f(x_0,t)|<\epsilon$ for all t when $|x-x_0|<\delta$; integrate. For the derivative, by the mean value theorem $\frac{f(x+h,t)-f(x,t)}{h}=\partial_x f(x+\theta h,t)$, which converges to $\partial_x f(x,t)$ uniformly in t by uniform continuity of $\partial_x f$; apply Proposition 6.5. ∎

Integrating ϕ over $[a,b]$ and using this result with $\Phi(y)=\int_c^d\int_a^y f(x,t)dx\,dt$ proves **Fubini's theorem for continuous functions**: $\int_a^b\int_c^d f\,dt\,dx=\int_c^d\int_a^b f\,dx\,dt$. (Differentiate both sides in the upper limit y.)

For improper integrals in t we need uniformity. $\int_c^\infty f(x,t)dt$ **converges uniformly** on A if $\sup_{x\in A}|\int_T^\infty f(x,t)dt|\to 0$ as $T\to\infty$.

Theorem 8.2 (Dominated improper integrals)

Let f be continuous on $[a,b]\times[c,\infty)$ with $|f(x,t)|\le g(t)$ and $\int_c^\infty g<\infty$. Then $\phi(x)=\int_c^\infty f(x,t)dt$ converges uniformly and is continuous on $[a,b]$.

Proof. $|\phi(x)-\int_c^T f(x,t)dt|\le\int_T^\infty g\to 0$ uniformly in x, and each truncated integral is continuous by Proposition 8.1; apply Theorem 6.1. ∎

Theorem 8.3 (Differentiation of improper integrals)

Let f and $\partial_x f$ be continuous on $(a,b)\times[c,\infty)$, $\int_c^\infty f(x_0,t)dt$ converge for some x_0, and $\int_c^\infty \partial_x f(x,t)dt$ converge uniformly on (a,b). Then $\phi(x)=\int_c^\infty f(x,t)dt$ converges, is differentiable, and $\phi'(x)=\int_c^\infty \partial_x f(x,t)dt$.

Proof. Apply Theorem 6.4 to $\phi_T(x)=\int_c^T f(x,t)dt$ as $T\to\infty$ (along any sequence), using Proposition 8.1 for ϕ_T'. ∎

Example 8.1

(a) The Gaussian integral: $I=\int_0^\infty e^{-t^2}dt=\frac{\sqrt{\pi}}{2}$. Let $F(x)=\left(\int_0^x e^{-t^2}dt\right)^2$ and $G(x)=\int_0^1\frac{e^{-x^2(1+t^2)}}{1+t^2}dt$. Differentiating (Proposition 8.1) and substituting $s=xt$ gives $F'+G'=0$, so $F+G=G(0)=\frac{\pi}{4}$; letting $x\to\infty$, $I^2=\frac{\pi}{4}$.
(b) The Dirichlet integral: $\phi(x)=\int_0^\infty e^{-xt}\frac{\sin t}{t}dt$ for $x>0$ has $\phi'(x)=-\int_0^\infty e^{-xt}\sin t\,dt=-\frac{1}{1+x^2}$ (the derivative integral converges uniformly on $x\ge\delta>0$), and $\phi(x)\to 0$ as $x\to\infty$, so $\phi(x)=\frac{\pi}{2}-\arctan x$. Uniform convergence on $[0,\infty)$ (via integration by parts) lets $x\to 0+$: $\int_0^\infty\frac{\sin t}{t}dt=\frac{\pi}{2}$.

8.2 The gamma function

Definition (Gamma and beta functions)

$\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}dt$ for $x > 0$, and $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$ for $x, y > 0$.

Both integrals converge (compare with t^{x-1} near 0 and $e^{-t/2}$ at infinity). By Theorems 8.2–8.3 on compact subintervals of $(0,\infty)$, Γ is C^∞ with $\Gamma^{(k)}(x) = \int_0^\infty (\log t)^k t^{x-1}e^{-t}dt$. Integration by parts gives $\Gamma(x+1) = x\Gamma(x)$, and $\Gamma(1) = 1$, so $\Gamma(n+1) = n!$. Also $\Gamma(\tfrac12) = 2\int_0^\infty e^{-s^2}ds = \sqrt{\pi}$.

A positive function is **log-convex** if $\log f$ is convex. By Hölder's inequality ($\int uv \le (\int u^p)^{1/p}(\int v^q)^{1/q}$, $\frac1p + \frac1q = 1$) applied to $t^{x/p-1/p}e^{-t/p}$ and $t^{y/q-1/q}e^{-t/q}$,

$$\Gamma\Big(\frac{x}{p} + \frac{y}{q}\Big) \le \Gamma(x)^{1/p}\Gamma(y)^{1/q},$$

so Γ is log-convex. A continuous function satisfying the midpoint inequality $f(\frac{x+y}{2}) \le \frac{f(x)+f(y)}{2}$ is convex (the inequality extends to dyadic weights by induction, and to all weights by continuity), which is often an easier way to verify convexity.

Theorem 8.4 (Bohr–Mollerup)

Γ is the only function $f : (0,\infty) \to (0,\infty)$ such that (a) $f(x+1) = xf(x)$, (b) $f(1) = 1$, (c) $\log f$ is convex. Moreover, for $x > 0$,

$$\Gamma(x) = \lim_{n\to\infty} \frac{n!\,n^x}{x(x+1)\cdots(x+n)}.$$

Proof. Let f satisfy (a)–(c) and $\varphi = \log f$. By (a), (b) it suffices to treat $0 < x < 1$. For integers n, convexity bounds the slope of φ on $[n, n+x]$ between the slopes on $[n-1, n]$ and $[n, n+1]$, which are $\log(n-1)$ and $\log n$:

$$x\log(n-1) \le \varphi(n+x) - \varphi(n) \le x\log n.$$

Since $\varphi(n) = \log(n-1)!$ and $\varphi(n+x) = \varphi(x) + \log\{x(x+1)\cdots(x+n-1)\}$, this gives $\frac{(n-1)!\,(n-1)^x}{x(x+1)\cdots(x+n-1)} \le f(x) \le \frac{(n-1)!\,n^x}{x(x+1)\cdots(x+n-1)}$. Both bounds tend to the stated limit (their ratio tends to 1), so f is determined, and Γ satisfies (a)–(c). ∎

Corollary 8.5

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

Proof. Fix y and let $f(x) = \Gamma(x+y)B(x,y)/\Gamma(y)$. Integration by parts gives $B(x+1,y) = \frac{x}{x+y}B(x,y)$, so $f(x+1) = xf(x)$; $B(1,y) = 1/y$ gives $f(1) = 1$; and $B(\cdot,y)$ is log-convex by Hölder, as is $\Gamma(\cdot + y)$. By Bohr–Mollerup, $f = \Gamma$. ∎

Example 8.2

(a) $\int_0^{\pi/2} \sin^{2x-1}\theta\cos^{2y-1}\theta\,d\theta = \frac12 B(x,y)$ (substitute $t = \sin^2\theta$); e.g. $\int_0^{\pi/2}\sin^n\theta\,d\theta = \frac{\sqrt{\pi}}{2}\frac{\Gamma((n+1)/2)}{\Gamma(n/2+1)}$.

(b) The volume of the unit ball in $\mathbb{R}^n$ is $\pi^{n/2}/\Gamma(\frac{n}{2}+1)$ (compute $\int_{\mathbb{R}^n} e^{-\|x\|^2}dx = \pi^{n/2}$ in two ways).

(c) Stirling's formula $\Gamma(x+1) \sim \sqrt{2\pi x}\, x^x e^{-x}$ follows by writing $\Gamma(x+1) = x^{x+1}e^{-x}\int_0^\infty e^{-x(s-1-\log s)}ds$ and noting that $s-1-\log s \approx (s-1)^2/2$ near $s=1$ (Laplace's method).

8.3 Integral transforms

A function of the form $F(x) = \int_I k(x,t)f(t)dt$ is an **integral transform** of f with **kernel** k. Gamma and beta are of this type with $k(x,t) = t^{x-1}$ (the Mellin transform), and the Hölder argument above shows that $\log F$ is convex whenever $f \geq 0$ and F is continuous.

The **Laplace transform** is $(\mathcal{L}f)(x) = \int_0^\infty e^{-xt}f(t)dt$, defined for x large when, e.g., $|f(t)| \leq Ce^{at}$. Direct computation gives

$$\mathcal{L}[e^{at}](x) = \frac{1}{x-a}, \quad \mathcal{L}[\cos at](x) = \frac{x}{x^2+a^2}, \quad \mathcal{L}[\sin at](x) = \frac{a}{x^2+a^2} \qquad (x > a,\ x > 0),$$

the last two from the first with a replaced by $\pm ia$. If f is C^1 and f, f' grow at most exponentially, integration by parts gives

$$\mathcal{L}[f'](x) = x\,\mathcal{L}[f](x) - f(0),$$

which turns linear differential equations with constant coefficients into algebraic equations.

The **Fourier transform** is $\hat{f}(x) = \int_{-\infty}^\infty e^{-itx}f(t)dt$, defined for every x when $\int|f| < \infty$. The Laplace transform is the special case of complex argument: with $x = u+iv$, $\int_0^\infty e^{-tx}f = \int_0^\infty e^{-itv}e^{-tu}f(t)dt$. By Theorem 8.2, $\hat{f}$ is continuous and $|\hat{f}| \leq \int|f|$; the Riemann–Lebesgue lemma of Chapter 9 shows $\hat{f}(x) \to 0$ as $|x| \to \infty$. For example, $f = \mathbf{1}_{[-1,1]}$ gives $\hat{f}(x) = \frac{2\sin x}{x}$, and $f(t) = e^{-t^2/2}$ gives $\hat{f}(x) = \sqrt{2\pi}e^{-x^2/2}$ (differentiate under the integral to get $\hat{f}' = -x\hat{f}$).

Chapter 9. Fourier Series

Throughout, functions are 2π-periodic and complex valued, $\mathcal{R}^1 = \mathcal{R}^1[-\pi,\pi]$, and

$$\hat f(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt, \qquad s_N(f;x) = \sum_{n=-N}^{N} \hat f(n)e^{inx}.$$

The **Fourier series** of f is $\sum_{n\in\mathbb{Z}} \hat f(n)e^{inx}$; in real form $\frac{a_0}{2} + \sum(a_n\cos nx + b_n \sin nx)$ with $a_n = \hat f(n) + \hat f(-n)$, $b_n = i(\hat f(n) - \hat f(-n))$.

9.1 Orthonormal systems

In a complex inner product space X, $\{e_n\}$ is **orthonormal** if $\langle e_m, e_n\rangle = \delta_{mn}$. With $\langle f,g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f\overline{g}$, the functions $e_n(x) = e^{inx}$ form an orthonormal system in $\mathcal{R}^2$, and $\hat f(n) = \langle f, e_n\rangle$.

Theorem 9.1 (Best approximation and Bessel's inequality)

For $v \in X$ and an orthonormal system $\{e_n\}$, put $c_n = \langle v, e_n\rangle$. Among all finite combinations $\sum_{|n|\le N} a_n e_n$, the distance to v is minimized exactly by $a_n = c_n$, and

$$\Big\| v - \sum_{|n|\le N} c_n e_n \Big\|^2 = \|v\|^2 - \sum_{|n|\le N} |c_n|^2.$$

Consequently $\sum_n |c_n|^2 \le \|v\|^2$ (Bessel). In particular $\sum|\hat f(n)|^2 \le \frac{1}{2\pi}\int|f|^2$ for $f \in \mathcal{R}^2$.

Proof. Expanding, $\|v - \sum a_n e_n\|^2 = \|v\|^2 - \sum|c_n|^2 + \sum|a_n - c_n|^2$. ∎

So $s_N(f)$ is the best L^2 approximation of f by trigonometric polynomials of degree N. Basic rules: $\widehat{f(\cdot - a)}(n) = e^{-ina}\hat f(n)$; $\widehat{f'}(n) = in\hat f(n)$ for C^1 periodic f (integration by parts); and the convolution $f * g(x) = \frac{1}{2\pi}\int f(x-t)g(t)dt$ satisfies $\widehat{f*g} = \hat f \hat g$.

Example 9.1

(a) $f(x) = x$ on $(-\pi,\pi)$: $\hat f(0) = 0$, $\hat f(n) = \frac{(-1)^n i}{n}$, so $f \sim 2\sum_{n\ge1}(-1)^{n+1}\frac{\sin nx}{n}$.
(b) $f(x) = |x|$ on $[-\pi,\pi]$: $f \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k\ge0}\frac{\cos(2k+1)x}{(2k+1)^2}$, converging uniformly by the M-test. At $x = 0$: $\sum \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$, hence $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$.

9.2 Pointwise convergence

Theorem 9.2 (Riemann–Lebesgue lemma)

If $\int_I |f| < \infty$ on an interval I (possibly infinite), then $\int_I f(t)e^{i\lambda t}dt \to 0$ as $|\lambda| \to \infty$. In particular $\hat f(n) \to 0$.

Proof. For the indicator of an interval $[\alpha,\beta]$ the integral is $\frac{e^{i\lambda\beta} - e^{i\lambda\alpha}}{i\lambda} \to 0$, hence the claim holds for step functions; for general f approximate in $\|\cdot\|_1$ by step functions (Theorem 7.9). ∎

The partial sums are convolutions with the **Dirichlet kernel**:

$$s_N(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_N(t)dt, \qquad D_N(t) = \sum_{|n|\le N} e^{int} = \frac{\sin(N+\frac12)t}{\sin\frac{t}{2}}, \qquad \frac{1}{2\pi}\int D_N = 1.$$

Theorem 9.3 (Localization)

If $f \in \mathcal{R}^1$ and $f = 0$ on an interval around x, then $s_N(f;x) \to 0$. Hence the convergence of $s_N(f;x)$ and its limit depend only on the values of f near x.

Proof. If $f(x-t) = 0$ for $|t| < \delta$, then $s_N(f;x) = \frac{1}{2\pi}\int_{\delta \le |t| \le \pi} \frac{f(x-t)}{\sin(t/2)} \sin(N+\frac{1}{2})t\,dt$, and $f(x-t)/\sin(t/2)$ is integrable there; apply Riemann–Lebesgue. ∎

Theorem 9.4 (Dini's test)

Let $f \in \mathcal{R}^1$ and $s \in \mathbb{C}$. If $\int_0^\delta \frac{|f(x+t)+f(x-t)-2s|}{t} dt < \infty$ for some $\delta > 0$, then $s_N(f;x) \to s$.

Proof. Using the evenness of D_N, $s_N(f;x) - s = \frac{1}{2\pi}\int_0^\pi \{f(x+t)+f(x-t)-2s\} D_N(t) dt$, and $\frac{f(x+t)+f(x-t)-2s}{\sin(t/2)}$ is integrable on $(0,\pi]$ by hypothesis (since $\sin(t/2) \ge t/\pi$). Riemann–Lebesgue finishes the proof. ∎

Corollary 9.5

If f satisfies a Lipschitz (or Hölder) condition at x, or is differentiable at x, then $s_N(f;x) \to f(x)$. If f has one-sided limits $f(x\pm)$ and one-sided derivatives at a jump, then $s_N(f;x) \to \frac{1}{2}\{f(x+)+f(x-)\}$.

Theorem 9.6 (Dirichlet–Jordan)

If f is of bounded variation on an interval around x, then $s_N(f;x) \to \frac{1}{2}\{f(x+)+f(x-)\}$; if f is continuous and BV on a closed interval, the convergence is uniform on compact subintervals of its interior.

Sketch. By localization and the Jordan decomposition one may assume $g(t) = f(x+t)+f(x-t)-f(x+)-f(x-)$ is increasing on $[0,\delta]$ with $g(0+) = 0$. By the second mean value theorem for integrals (Chapter 5), $\int_0^\eta g(t) D_N(t) dt = g(\eta-)\int_c^\eta D_N$ for some c, and $|\int_c^\eta D_N|$ is bounded independently of N, c, η (because $\int_0^y \frac{\sin u}{u} du$ is bounded). Choosing η with $g(\eta-)$ small and applying Riemann–Lebesgue on $[\eta,\pi]$ gives the result. ∎

Example 9.2

(a) For $f(x) = x$ on $(-\pi,\pi)$, Dini's test gives $x = 2\sum(-1)^{n+1}\frac{\sin nx}{n}$ on $(-\pi,\pi)$, and the series is $0 = \frac{1}{2}(f(\pi-)+f(-\pi+))$ at $\pm\pi$. Setting $x = \frac{\pi}{2}$ gives Leibniz's series for $\frac{\pi}{4}$.
(b) Continuity alone does not guarantee convergence: there are continuous functions whose Fourier series diverge at a point (du Bois-Reymond), essentially because $\frac{1}{2\pi}\int|D_N| \sim \frac{4}{\pi^2}\log N \to \infty$.
(c) Near a jump, s_N overshoots by about 9% of the jump no matter how large N is (Gibbs phenomenon), so the convergence in Corollary 9.5 is not uniform near jumps.

9.3 Fejér's theorem and Parseval's identity

The Cesàro means $\sigma_N = \frac{1}{N+1}\sum_{k=0}^N s_k$ are convolutions with the **Fejér kernel**

$$K_N(t) = \frac{1}{N+1}\sum_{k=0}^{N} D_k(t) = \frac{1}{N+1}\Big(\frac{\sin\frac{N+1}{2}t}{\sin\frac{t}{2}}\Big)^2,$$

which, unlike D_N, is nonnegative, has $\frac{1}{2\pi}\int K_N = 1$, and satisfies $\sup_{\delta\le|t|\le\pi} K_N(t) \to 0$ for every $\delta > 0$ (an approximate identity).

Theorem 9.7 (Fejér)

If $f \in \mathcal{R}^1$ has one-sided limits at x, then $\sigma_N(f;x) \to \frac{1}{2}\{f(x+) + f(x-)\}$. If f is continuous (and periodic), $\sigma_N(f) \to f$ uniformly.

Proof. $\sigma_N(f;x) - s = \frac{1}{2\pi}\int_0^\pi \{f(x+t) + f(x-t) - 2s\}K_N(t)dt$. Split at δ: on $[0,\delta]$ the bracket is small and $K_N \ge 0$ has integral at most 1; on $[\delta,\pi]$, $K_N \to 0$ uniformly and f is integrable. For continuous f, uniform continuity makes δ independent of x. ∎

Corollary 9.8

(a) Trigonometric polynomials are uniformly dense in the continuous periodic functions. (b) If the Fourier series of a continuous periodic f converges at x, its sum is $f(x)$. (c) If $\hat f(n) = 0$ for all n and f is continuous, then $f = 0$.

Proof. (a) $\sigma_N(f)$ is a trigonometric polynomial. (b) If $s_N \to s$, then $\sigma_N \to s$ (Cesàro), while $\sigma_N \to f(x)$. (c) $\sigma_N(f) = 0$. ∎

Theorem 9.9 (Parseval's identity)

For $f, g \in \mathcal{R}^2$: $\|f - s_N(f)\|_2 \to 0$,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f|^2 = \sum_{n\in\mathbb{Z}} |\hat f(n)|^2, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} f\overline{g} = \sum_n \hat f(n)\overline{\hat g(n)}.$$

Proof. Given ϵ, choose continuous periodic h with $\|f - h\|_2 < \epsilon$ (Theorem 7.9) and a trigonometric polynomial P of degree M with $\|h - P\|_\infty < \epsilon$ (Corollary 9.8). For $N \ge M$, by best approximation, $\|f - s_N(f)\|_2 \le \|f - P\|_2 < 2\epsilon$. The identity follows from Theorem 9.1, and the polarized form from the polarization identity. ∎

Example 9.3

Parseval for $f(x) = x$: $\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{3} = \sum_{n\ne 0}\frac{1}{n^2}$, again $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$. For $f(x) = x^2$ one obtains $\sum \frac{1}{n^4} = \frac{\pi^4}{90}$.

9.4 Smoothness, decay, differentiation and integration

Theorem 9.10 (Term-by-term integration)

For $f \in \mathcal{R}^2$ and any a, x, $\int_a^x f = \sum_n \hat f(n)\int_a^x e^{int}dt$, the series converging uniformly in x; this holds even if the Fourier series of f itself diverges.

Proof. $\left|\int_a^x (f - s_N(f))\right| \le \sqrt{2\pi}\cdot\sqrt{2\pi}\|f - s_N(f)\|_2 \to 0$ by Cauchy–Schwarz and Parseval. ∎

Smoothness of f corresponds to decay of $\hat f$. If $f \in C^k$ (periodic) then $\widehat{f^{(k)}}(n) = (in)^k \hat f(n)$, so $|\hat f(n)| = o(|n|^{-k})$ by Riemann–Lebesgue. If f is C^1, then $\sum|\hat f(n)| \le |\hat f(0)| + \left(\sum \frac{1}{n^2}\right)^{1/2}\left(\sum |n\hat f(n)|^2\right)^{1/2} < \infty$ and the Fourier series converges absolutely and uniformly to f.

Theorem 9.11

A periodic f is C^∞ iff $\hat f$ is rapidly decreasing ($|n|^k \hat f(n) \to 0$ for every k). Conversely every rapidly decreasing $a : \mathbb{Z} \to \mathbb{C}$ is $\hat f$ for the C^∞ function $f(x) = \sum a(n) e^{inx}$. A periodic f is real analytic iff $\limsup |\hat f(n)|^{1/|n|} < 1$.

Proof. The first statements follow from the paragraph above and Theorem 6.4 (the differentiated series converge uniformly). For analytic f, f extends holomorphically to a strip $|\operatorname{Im} z| < r$, and shifting the contour in $\hat f(n)$ gives $|\hat f(n)| \le C e^{-r'|n|}$ for $r' < r$; conversely geometric decay makes $\sum \hat f(n) e^{inz}$ converge in a strip. ∎

Not every null sequence is a Fourier coefficient sequence of an $\mathcal{R}^1$ (or even L^1) function: if f is odd and integrable, $\sum_{n \ge 1} \frac{\hat f(n)}{n}$ converges (it equals, up to a constant, $\int_0^\pi$ of the integral of f), whereas for $b_n = 1/\log n$ the series $\sum \frac{b_n}{n}$ diverges. So $\sum \frac{\sin nx}{\log n}$, which converges everywhere by Theorem 6.13, is not the Fourier series of an integrable function. Chapter 10 completes the picture with L^1, L^2 and the Riesz–Fischer theorem.

Chapter 10. The Lebesgue Integral

The Riemann integral handles continuous functions well but behaves badly under limits: pointwise limits of integrable functions need not be integrable, and the spaces $\mathcal{R}^1$, $\mathcal{R}^2$ are not complete (Example 7.2). Lebesgue's integral repairs both defects. We construct it on $\mathbb{R}$.

10.1 Measurable sets and measure

Proposition 10.1

Every open set $U \subset \mathbb{R}$ is a disjoint union of countably many open intervals, uniquely.

Proof. For $x \in U$ let I_x be the union of all open intervals containing x and contained in U; it is an open interval, and two such are equal or disjoint. Each contains a rational, so there are countably many. ∎

For an open set $U = \bigsqcup I_k$ put $m(U) = \sum |I_k|$, and for any $E \subset \mathbb{R}$ define the **outer measure**

$$m^*(E) = \inf\{m(U) : U \supset E \text{ open}\}.$$

Proposition 10.2

m^* is monotone, countably subadditive ($m^*(\bigcup E_n) \le \sum m^*(E_n)$), translation invariant, and $m^*(I) = |I|$ for intervals. Countable sets have outer measure 0.

Proof. Subadditivity: choose open $U_n \supset E_n$ with $m(U_n) < m^*(E_n) + \epsilon 2^{-n}$; then $\bigcup U_n$ is open and $m(\bigcup U_n) \le \sum m(U_n)$ (decompose the union into intervals and use compactness of closed subintervals). $m^*([a,b]) \ge b - a$ follows from the Heine–Borel theorem: an open cover has a finite subcover whose lengths sum to at least $b - a$. ∎

Definition (Lebesgue measurable set)

$E \subset \mathbb{R}$ is **measurable** if for every $\epsilon > 0$ there is an open $U \supset E$ with $m^*(U \setminus E) < \epsilon$. Then $m(E) = m^*(E)$. The class of measurable sets is denoted $\mathfrak{M}$.

Theorem 10.3

$\mathfrak{M}$ is a σ-algebra: $\emptyset \in \mathfrak{M}$; $E \in \mathfrak{M} \Rightarrow E^c \in \mathfrak{M}$; $E_n \in \mathfrak{M} \Rightarrow \bigcup E_n \in \mathfrak{M}$. It contains all open and closed sets and all sets of outer measure 0.

Sketch. Countable unions: choose $U_n \supset E_n$ with $m^*(U_n \setminus E_n) < \epsilon 2^{-n}$ and use subadditivity. Null sets: trivially measurable. Closed sets: a closed F is a countable union of compact sets, and for compact K and open $U \supset K$, $U \setminus K$ is open with $m(U \setminus K) = m(U) - m(K)$ (a computation with intervals). Complements: given E, choose open $U_n \supset E$ with $m^*(U_n \setminus E) < 1/n$; then $E^c = (\bigcup U_n^c) \cup Z$ with $Z = E^c \setminus \bigcup U_n^c \subset U_n \setminus E$ for all n, a null set, and $\bigcup U_n^c$ is a union of closed sets. ∎

Theorem 10.4 (Countable additivity)

If $E_n \in \mathfrak{M}$ are disjoint, then $m(\bigcup E_n) = \sum m(E_n)$.

Proof. $\le$ is subadditivity. For $\ge$, first for compact disjoint sets (they have positive distance, and an open cover of the union can be split), then for bounded measurable sets via inner approximation

by compact sets (E measurable implies for each ϵ a closed $F \subset E$ with $m(E \setminus F) < \epsilon$), and finally in general. ■

Consequently m is continuous along increasing sequences ($m(\bigcup E_n) = \lim m(E_n)$) and along decreasing sequences of finite measure. A property holds **almost everywhere** (a.e.) if it fails only on a null set. Not every set is measurable (Vitali's construction using a choice of representatives of $\mathbb{R}/\mathbb{Q}$), but every set that arises in practice is.

10.2 Measurable functions and the integral

Let $E \in \mathfrak{M}$ and $\mathbb{R}^* = [-\infty, \infty]$.

Proposition 10.5

For $f : E \to \mathbb{R}^*$ the following are equivalent, and define **measurability** of f: $\{f > a\} \in \mathfrak{M}$ for all $a \in \mathbb{R}$; $\{f \geq a\} \in \mathfrak{M}$ for all a; $\{f < a\} \in \mathfrak{M}$ for all a; $\{f \leq a\} \in \mathfrak{M}$ for all a.

Proof. $\{f \geq a\} = \bigcap_n \{f > a - \frac{1}{n}\}$, $\{f < a\} = \{f \geq a\}^c$, and so on. ■

Continuous functions, monotone functions, sums, products, max, min, $\sup_n f_n$, $\limsup f_n$ and pointwise limits of measurable functions are measurable; changing a measurable function on a null set keeps it measurable. A **simple function** is a finite combination $s = \sum c_i \mathbf{1}_{A_i}$ of indicators of measurable sets.

Proposition 10.6

Every measurable $f : E \to [0, \infty]$ is the pointwise limit of an increasing sequence of nonnegative simple functions.

Proof. $s_n = \sum_{k=1}^{n2^n} \frac{k-1}{2^n} \mathbf{1}\{\frac{k-1}{2^n} \leq f < \frac{k}{2^n}\} + n\mathbf{1}\{f \geq n\}$. ■

Definition (Lebesgue integral)

For simple $s = \sum c_i \mathbf{1}_{A_i} \geq 0$, $\int_E s = \sum c_i m(A_i \cap E)$. For measurable $f \geq 0$, $\int_E f = \sup\{\int_E s : 0 \leq s \leq f,\ s \text{ simple}\} \in [0, \infty]$. For real f with $\int |f| < \infty$ (written $f \in L^1(E)$), $\int f = \int f^+ - \int f^-$; for complex f, integrate real and imaginary parts.

Theorem 10.7

If $f \geq 0$ is measurable on $E = \bigsqcup A_n$ (disjoint measurable), then $\int_E f = \sum_n \int_{A_n} f$. If $f \geq 0$ and $\int_E f = 0$, then $f = 0$ a.e.

Proof. For simple f this is countable additivity of m; in general take suprema. For the second claim, $m\{f > 1/n\} \leq n \int f = 0$ (Chebyshev). ■

10.3 Convergence theorems

Theorem 10.8 (Monotone convergence)

If $0 \leq f_1 \leq f_2 \leq \cdots$ are measurable and $f_n \to f$ pointwise, then $\int f_n \to \int f$.

Proof. Clearly $\lim \int f_n \leq \int f$. For simple $0 \leq s \leq f$ and $c < 1$, the sets $E_n = \{f_n \geq cs\}$ increase to E, so $\int f_n \geq \int_{E_n} f_n \geq c \int_{E_n} s \to c \int s$ (Theorem 10.7). Let $c \to 1$ and take the supremum over s. ■

Consequences: the integral is additive on nonnegative functions (approximate by simple functions), and $\int \sum f_n = \sum \int f_n$ for $f_n \geq 0$.

Theorem 10.9 (Riemann versus Lebesgue)

A bounded $f : [a,b] \to \mathbb{R}$ is Riemann integrable iff the set of its discontinuities has measure zero, and then f is Lebesgue integrable with the same integral. Absolutely convergent improper Riemann integrals are Lebesgue integrals.

Sketch. Take partitions P_k refining each other with mesh $\to 0$; the step functions l_k, u_k built from m_i, M_i increase/decrease to limits $l \leq f \leq u$ with $l = u$ exactly at the continuity points of f (off the countable set of partition points). By monotone convergence $\int l = \lim L(f, P_k)$ and $\int u = \lim U(f, P_k)$, so f is Riemann integrable iff $\int (u - l) = 0$ iff $u = l$ a.e. ■

So the Dirichlet function, not Riemann integrable, has Lebesgue integral 0.

Theorem 10.10 (Fatou's lemma)

For measurable $f_n \geq 0$, $\int \liminf f_n \leq \liminf \int f_n$.

Proof. Apply monotone convergence to $g_n = \inf_{k \geq n} f_k \uparrow \liminf f_n$, noting $\int g_n \leq \int f_k$ for $k \geq n$. ■

Theorem 10.11 (Dominated convergence)

If $f_n \to f$ a.e., $|f_n| \leq g$ and $g \in L^1(E)$, then $f \in L^1$ and $\int |f_n - f| \to 0$; in particular $\int f_n \to \int f$.

Proof. Apply Fatou to $2g - |f_n - f| \geq 0$: $\int 2g \leq \liminf \int (2g - |f_n - f|) = \int 2g - \limsup \int |f_n - f|$. ■

Theorem 10.12 (Beppo Levi)

If $f_n \in L^1(E)$ and $\sum \int |f_n| < \infty$, then $\sum f_n$ converges absolutely a.e., its sum is in L^1, and $\int \sum f_n = \sum \int f_n$.

Proof. $g = \sum |f_n|$ has $\int g = \sum \int |f_n| < \infty$ by monotone convergence, so $g < \infty$ a.e.; apply dominated convergence to the partial sums, dominated by g. ■

Example 10.1

(a) Dominated convergence explains Example 6.1(b): $nx(1-x^2)^n$ has no integrable majorant.
(b) $\int_0^\infty \frac{x}{e^x - 1} dx = \sum_{n \geq 1} \int_0^\infty x e^{-nx} dx = \sum \frac{1}{n^2} = \frac{\pi^2}{6}$, by monotone convergence.
(c) Differentiation under the integral sign holds whenever $|\partial_x f(x,t)| \leq g(t)$ with $g \in L^1$ (mean value theorem plus dominated convergence), a much more flexible version of Theorem 8.3.

Theorem 10.13 (Absolute continuity)

If $f \in L^1(E)$, then for every $\epsilon > 0$ there is $\delta > 0$ with $\int_A |f| < \epsilon$ whenever $A \subset E$ is measurable with $m(A) < \delta$.

Proof. $\int |f| \mathbf{1}\{|f| > n\} \to 0$ by dominated convergence; choose n with this $< \epsilon/2$ and $\delta = \epsilon/(2n)$. ■

10.4 The spaces L^1 and L^2; Fourier series again

Identify functions equal a.e. Then $\|f\|_1 = \int |f|$ is a norm on $L^1(E)$, and $\|f\|_2 = (\int |f|^2)^{1/2}$ on $L^2(E)$, with inner product $\int f\bar{g}$.

Theorem 10.14 (Completeness)

$L^1(E)$ and $L^2(E)$ are complete; moreover every Cauchy sequence has a subsequence converging a.e.

Proof. By Proposition 7.8 it suffices that absolutely convergent series converge. For L^1 this is Beppo Levi's theorem. For L^2, if $\sum \|f_n\|_2 < \infty$, then $g = \sum |f_n|$ satisfies $\|g\|_2 \le \sum \|f_n\|_2$ (Minkowski plus monotone convergence), so $\sum f_n$ converges a.e. and, dominated by g^2, in L^2. ■

For periodic functions the theory of Chapter 9 extends verbatim from $\mathcal{R}^1, \mathcal{R}^2$ to L^1, L^2 on $[-\pi, \pi]$: Riemann–Lebesgue, Fejér's theorem in the L^1 norm, and Parseval's identity. Completeness of L^2 adds the converse:

Theorem 10.15 (Riesz–Fischer)

For every $c \in \ell^2(\mathbb{Z})$ there is $f \in L^2[-\pi, \pi]$ with $\hat{f}(n) = c(n)$ for all n. Thus $f \mapsto \hat{f}$ is an isometric isomorphism of $L^2[-\pi, \pi]$ (normalized measure) onto $\ell^2(\mathbb{Z})$.

Proof. The partial sums $P_N = \sum_{|n| \le N} c(n) e^{inx}$ form a Cauchy sequence in L^2 (orthonormality), so they converge to some f; then $\hat{f}(n) = \lim_N \langle P_N, e_n \rangle = c(n)$ by continuity of the inner product. Injectivity is Parseval. ■

Theorem 10.16 (Uniqueness)

If $f \in L^1[-\pi, \pi]$ and $\hat{f}(n) = 0$ for all n, then $f = 0$ a.e.

Proof. Fejér means $\sigma_N(f) = 0$ converge to f in L^1 (approximate f by continuous functions and use $\|\sigma_N(g)\|_1 \le \|g\|_1$, since $K_N \ge 0$ has integral 1). ■

Theorem 10.17 (Density)

Continuous functions with compact support (and step functions) are dense in $L^1(E)$ and in $L^2[a, b]$.

Proof. Simple functions are dense (Proposition 10.6 and dominated convergence); a measurable set of finite measure is approximated in measure by an open set, hence by a finite union of intervals, whose indicator is approximated by continuous functions. ■

10.5 Function spaces and Fourier coefficients

The correspondence $f \mapsto \hat{f}$ links function spaces with sequence spaces:

analytic periodic functions	$\leftrightarrow$	$\{a : \limsup \lvert a(n)\rvert^{1/\lvert n\rvert} < 1\}$
C^∞ periodic functions	$\leftrightarrow$	rapidly decreasing sequences
continuous periodic functions	$\to$	(a proper subset of) $\ell^2(\mathbb{Z})$
$L^2[-\pi, \pi]$	$\leftrightarrow$	$\ell^2(\mathbb{Z})$
$L^1[-\pi, \pi]$	$\to$	(a proper subset of) $c_0(\mathbb{Z})$

The map $L^1 \to c_0$ is injective (Theorem 10.16) but not onto (Section 9.4). Moreover, the coefficients of an L^1 function need not be square-summable, as the following shows.

Example 10.2 (Periodization)

For $g \in L^1(\mathbb{R})$, $G(x) = \sum_{n\in\mathbb{Z}} g(x - 2n\pi)$ converges a.e. and defines $G \in L^1[-\pi, \pi]$ with $\int_{-\pi}^{\pi}|G| \le \int_{\mathbb{R}}|g|$ (monotone convergence), and dominated convergence gives

$$\hat{G}(n) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(x)e^{-inx}dx = \frac{1}{2\pi}\hat{g}(n).$$

Choosing $g \in L^1(\mathbb{R})$ whose Fourier transform is not square-summable on $\mathbb{Z}$ (e.g. $g(x) = |x|^{-1/2}\mathbf{1}\{|x| \le 1\}$, for which $\hat{g}(n) \sim c|n|^{-1/2}$) produces $G \in L^1$ with $\hat{G} \notin \ell^2$. The same identity, combined with Fourier inversion, yields the Poisson summation formula $\sum g(2n\pi) = \frac{1}{2\pi}\sum \hat{g}(n)$ for nice g.